Islamic University of Technology

LAB MINI PROJECT IDEA SUBMISSION

COURSE CODE: EEE 4204

COURSE TITLE: ELECTRONICS I LAB (HW)

TEAM MEMBERS

1. SHIHAB SHAHRIAR SAMI – 240021223
2. YASIR ARAFAT – 240021202
3. ISHMAM DIBYA – 240021204
4. ISTIAK SARKER - 240021206
5. MAISHA AHMED RAIKA – 240021209
6. ASHJAI ARABI – 240021227

SMART LASER ALARM with DUAL LDR INTRUDER DETECTION

Description:

This project is a laser-based security alarm that detects when someone crosses a protected area. A laser beam is aimed across the area onto two LDR sensors placed a short distance apart. As long as the beam stays uninterrupted, the circuit remains idle. When an intruder blocks the beam, the sudden change in light triggers a 555 timer circuit, which sounds an alarm for a short, fixed duration and then automatically resets itself, ready to detect the next intrusion. Using two LDRs instead of one helps the system ignore small animals or debris and respond only to larger objects like a human.

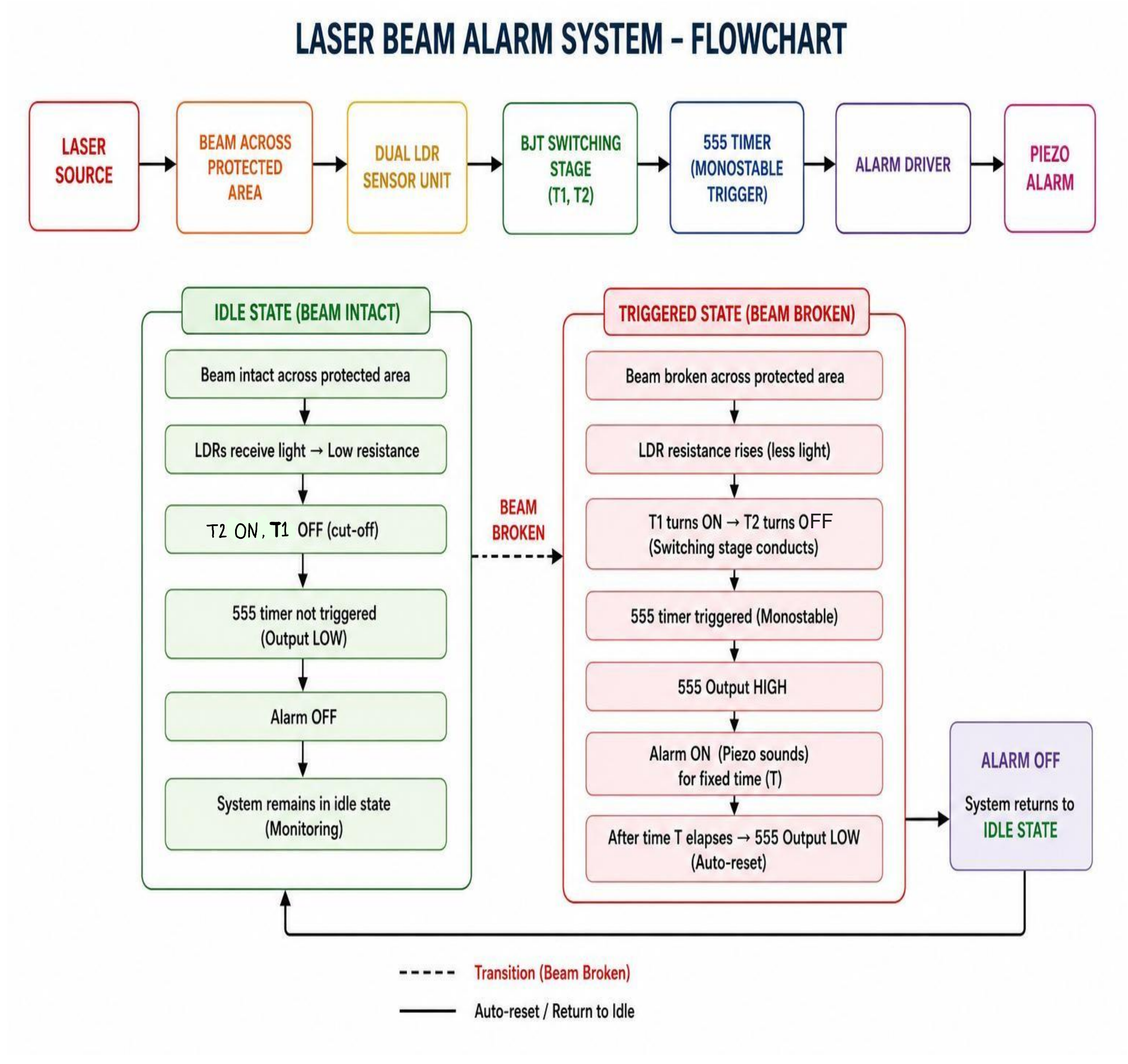
Proposed System Overview:

A laser beam is aligned across the protected zone so that it falls continuously on two LDR sensors housed in opaque tubes. As long as the beam is uninterrupted, both LDRs remain lowresistance, holding the sensing transistor off and keeping the system idle. When an intruder crosses the beam, blocking it, the LDR resistance rises, switching a chain of two BJTs that pulls the trigger pin of a 555 timer IC low. The 555 then activates an alarm for a preset time (a few seconds to minutes) before automatically resetting and re-arming, ready to detect the next intrusion.

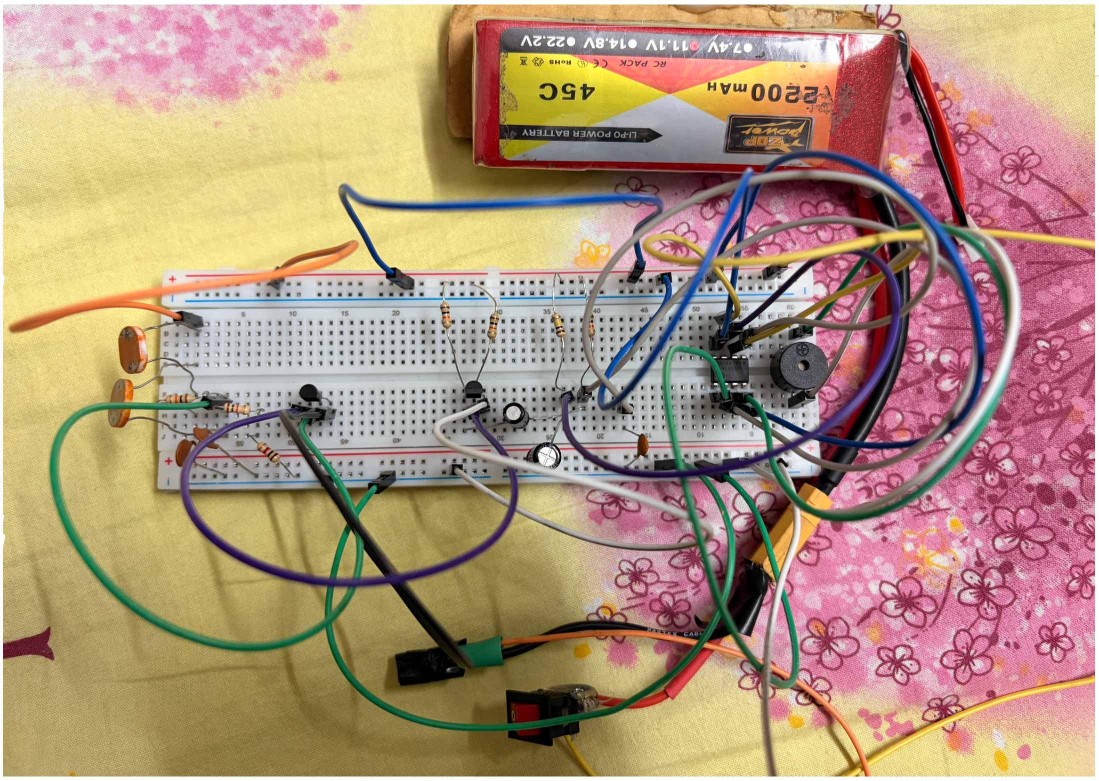
Two BJTs form the light-to-trigger interface between the LDR sensors and the 555 timer. T2 (NPN) acts as a light-controlled switch — the LDRs' resistance directly sets its base bias, so it stays ON while the laser beam is unbroken and turns OFF when the beam is interrupted. T1 (NPN) acts as a trigger driver — it's switched ON by T2's output and, in turn, pulls the 555's trigger pin (pin 2) low through a coupling capacitor, initiating the timed alarm output. This two-transistor arrangement isolates

the sensitive LDR sensing stage from the 555's trigger input and gives clean, reliable switching action.

Flowchart:

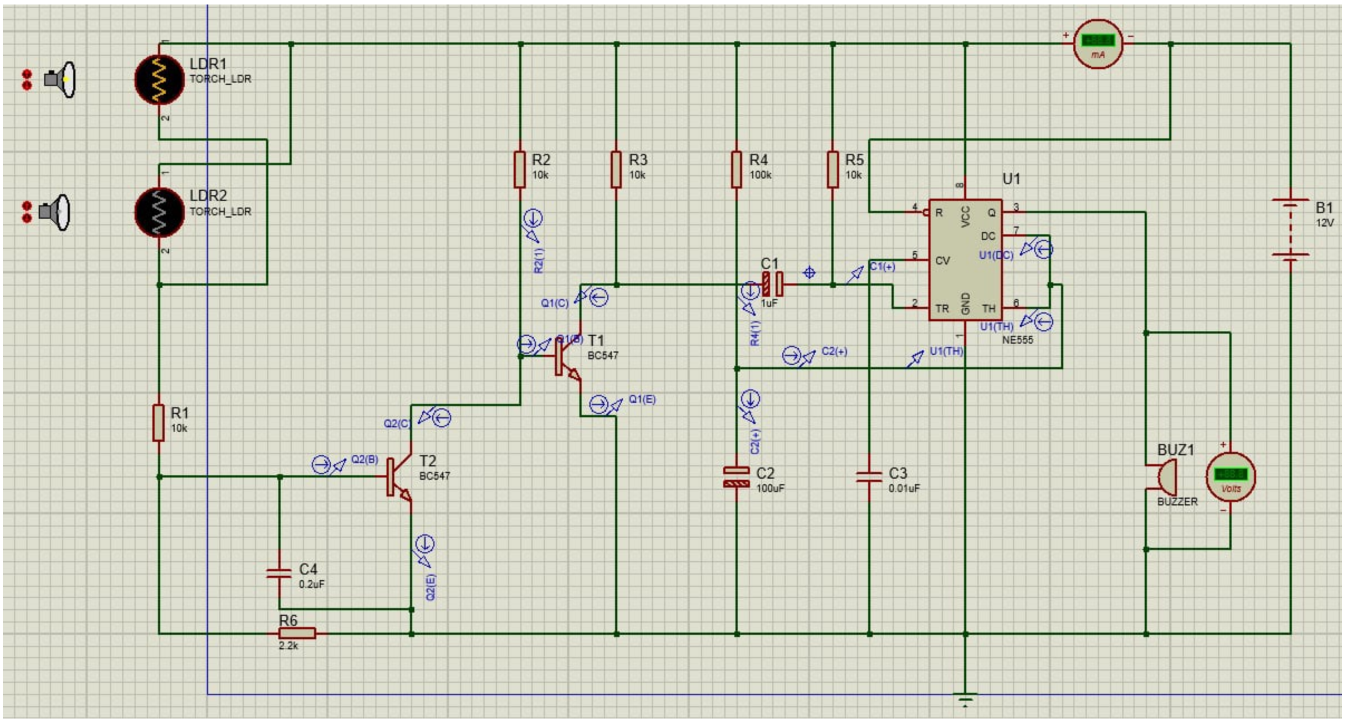

LASER BEAM ALARM SYSTEM - FLOWCHART
LASER SOURCE
BEAM ACROSS PROTECTED AREA
DUAL LDR SENSOR UNIT
BJT SWITCHING STAGE (T1, T2)
555 TIMER (MONOSTABLE TRIGGER)
ALARM DRIVER
PIEZO ALARM
IDLE STATE (BEAM INTACT)
Beam intact across protected area
LDRs receive light → Low resistance
T2 ON, T1 OFF (cut-off)
555 timer not triggered (Output LOW)
Alarm OFF
System remains in idle state (Monitoring)
BEAM BROKEN
TRIGGERED STATE (BEAM BROKEN)
Beam broken across protected area
LDR resistance rises (less light)
T1 turns ON → T2 turns OFF (Switching stage conducts)
555 timer triggered (Monostable)
555 Output HIGH
Alarm ON (Piezo sounds) for fixed time (T)
After time T elapses → 555 Output LOW (Auto-reset)
ALARM OFF
System returns to IDLE STATE
Transition (Beam Broken)
Auto-reset / Return to Idle

Circuit Diagram:



Simulation :

The circuit is simulated using Proteus and gives the desired outcome,



LDR1
TORCH_LDR
LDR2
TORCH_LDR
R1
10k
R2
10k
R3
10k
R4
100k
R5
10k
R2(1)
R4(1)
Q1(C)
Q1(B)
Q1(E)
T1
BC547
Q2(C)
Q2(B)
Q2(E)
T2
BC547
C1
1uF
C1(+)
C2(+)
C2
100uF
C3
0.01uF
C4
0.2uF
R6
2.2k
U1
NE555
R
VCC
Q
DC
CV
TR
GND
TH
U1(DC)
U1(TH)
4
8
3
7
5
2
6
1
mA
B1
12V
BUZ1
BUZZER
Volts

Fig: Simulation of Smart laser alarm circuit

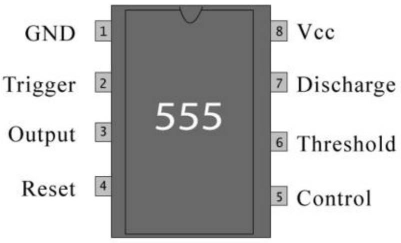
Components Required:

Component	**Value/Type**
IC	555 Timer
T1	BC547 (or PNP equivalent)
T2	BC547
LDR	2x, standard
R1, R2, R3, R5	10 KΩ
R4	100 KΩ
C1	1 µF / 25V
C2	100 µF / 25V
C3	0.01 µF
Alarm	12V, 200 mA piezo buzzer
Laser module	Standard laser pointer/diode
Power supply	9–12V DC
Breadboard, connecting wires	—

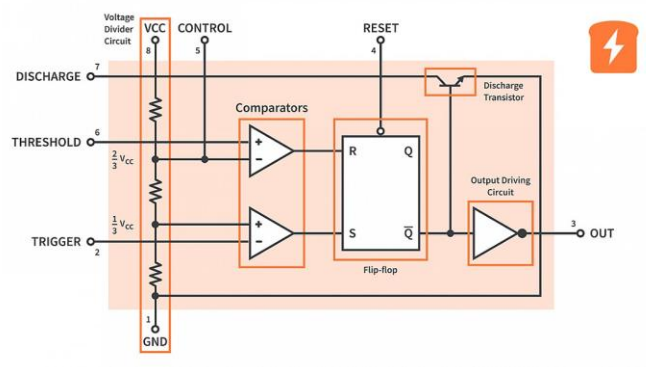
Working principal of IC 555 timer :

In this project, 555 timer is operating in Monostable Multivibrator Configuration.

Figure:


GND
1
Trigger
2
Output
3
Reset
4
555
8
Vcc
7
Discharge
6
Threshold
5
Control

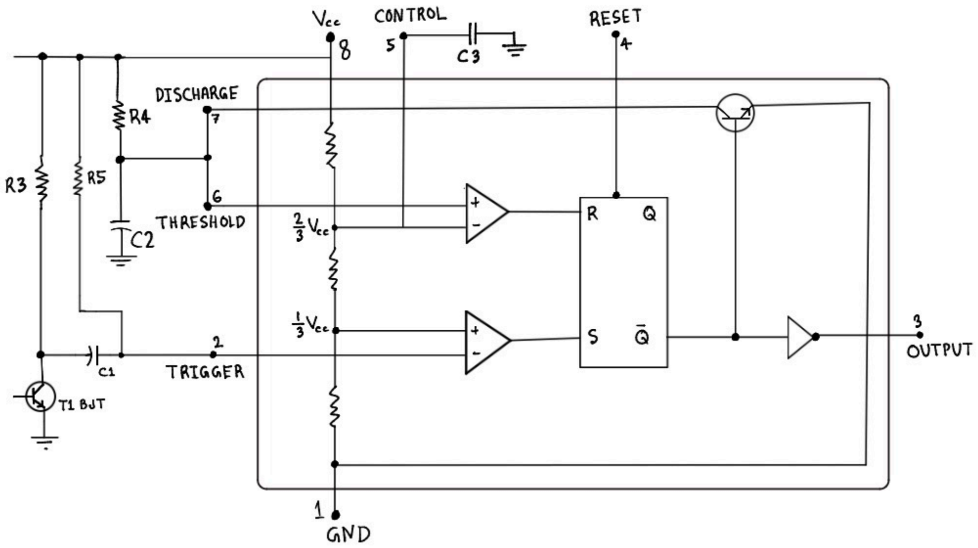
Circuit Diagram:


Voltage Divider Circuit
VCC
8
CONTROL
5
RESET
4
DISCHARGE
7
Discharge Transistor
Comparators
THRESHOLD
6
2/3 Vcc
R
Q
Output Driving Circuit
1/3 Vcc
TRIGGER
2
S
Q̄
3
OUT
Flip-flop
1
GND

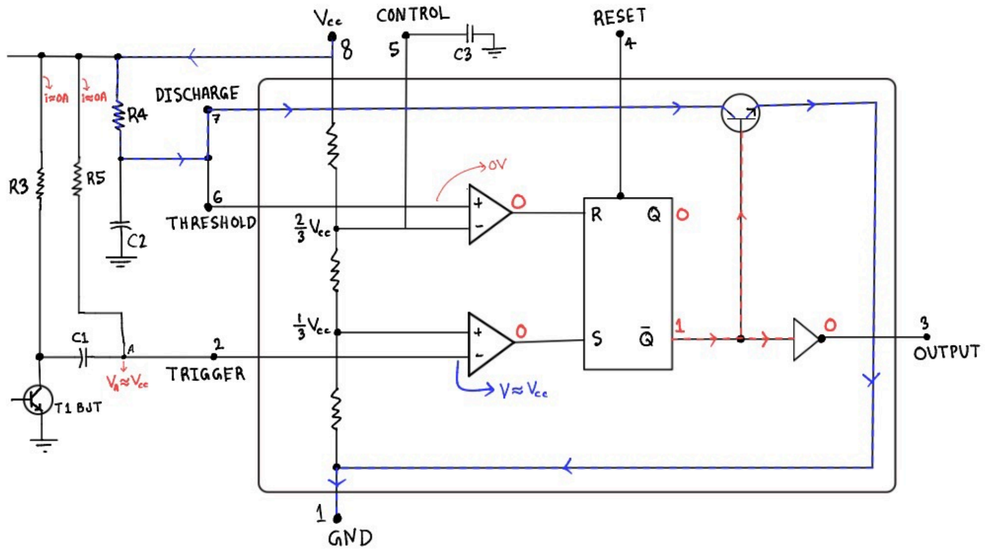
Truth Table of SR Flip-Flop:

S	R	Q	$\bar{Q}$
0	0	0	1
1	0	1	0
0	1	0	1
1	1	Invalid	Invalid

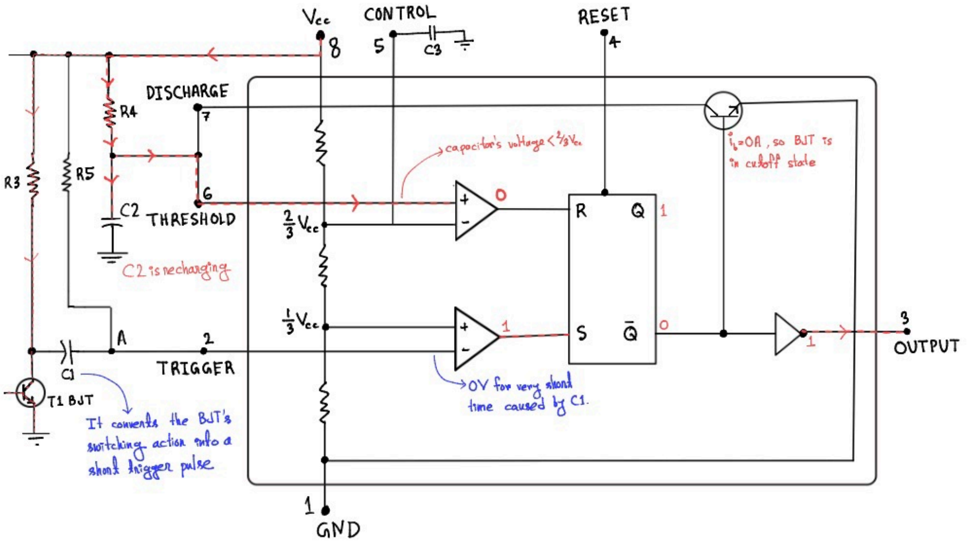
When IC 555 Timer is operating in Monostable Multivibrator Configuration,


Vcc
8
CONTROL
5
C3
RESET
4
DISCHARGE
7
R4
R3
R5
6
THRESHOLD
C2
2/3 Vcc
R
Q
1/3 Vcc
2
TRIGGER
C1
T1 BJT
S
Q̄
3
OUTPUT
1
GND

When T1 BJT is in cutoff mode:


Vcc
8
CONTROL
5
C3
RESET
4
i≈0A
i≈0A
R4
DISCHARGE
7
R3
R5
6
C2
THRESHOLD
$\frac{2}{3}V_{cc}$
0V
0
R
Q
0
$\frac{1}{3}V_{cc}$
0
S
$\bar{Q}$
1
0
3
OUTPUT
C1
A
2
$V_A \approx V_{cc}$
TRIGGER
$V \approx V_{cc}$
T1 BJT
1
GND

When T1 BJT is in saturation mode:


Vcc
8
CONTROL
5
C3
RESET
4
DISCHARGE
R4
7
R3
R5
6
C2
THRESHOLD
$\frac{2}{3}V_{cc}$
0
R
Q
1
$\frac{1}{3}V_{cc}$
1
S
$\bar{Q}$
0
1
3
OUTPUT
A
2
C1
TRIGGER
T1 BJT
1
GND

capacitor's voltage < $\frac{2}{3}V_{cc}$

i_b=0A , so BJT is in cutoff state

C2 is recharging

It converts the BJT's switching action into a short trigger pulse

0V for very short time caused by C1.

As C2 is recharging, output will be 1 until C2's voltage reaches 2/3Vcc. When C2's voltage is higher than 2/3Vcc, R will be 1, making the output low or 0.

Capacitors:

1) C1 = 1 μF / 25 V: trigger coupling capacitor

C1 is connected between the transistor sensing stage and **pin 2, TRIGGER, of the 555 timer**.

Its main purpose is to convert the transistor's switching action into a **short trigger pulse**.

When the laser beam is interrupted, the transistor stage changes voltage suddenly. C1 passes this sudden change to pin 2 of the 555, causing pin 2 to momentarily fall low. For a 555 timer:
VTRIG< ⅓VCC
causes the timer to trigger.

VTRIG is the voltage at **555 pin 2 (Trigger)** relative to **pin 1 (Ground)**. As Vpin1 is grounded so VTRIG is equal to Vpin2. is the power-supply voltage between **pin 8 and pin 1** of the 555.

In the circuit:

- Pin 8 → to
- Pin 1 → Ground

So if we use a 12 V supply:

VCC=12−0=12V Then the trigger

threshold is: 1/3VCC=1/3(12)=4V

Therefore, the 555 triggers when:

VTRIG<4V

The important thing is that C1 passes the **change in voltage**, but it does not continuously pass DC. So instead of holding pin 2 LOW for the entire time the laser is blocked, it gives the 555 a short triggering pulse.

This is useful because you want:

Beam broken→one trigger pulse→alarm for fixed time

rather than keeping the 555 permanently triggered.

In short:

C1 is a coupling or trigger capacitor. It converts the switching action of the transistor circuit into a short pulse for triggering pin 2 of the 555 timer.

R5 helps return pin 2 to its normal HIGH state after the pulse.

2) C2 = 100 µF / 25 V: main timing capacitor

This is the most important capacitor for the **555 monostable operation**.

C2 is connected to the **threshold/discharge network around pins 6 and 7**. Together with R4. it determines **how long the alarm remains ON**.

For a 555 monostable: T=1.1RC

Therefore:

R=1000Kohm

C=100uF

T=11s

So after someone breaks the laser beam, the alarm should remain active for approximately **11 seconds**, depending on component tolerances.

Immediately after triggering, C2 starts charging

through R4: 0V->Vc When the capacitor voltage

reaches approximately: VC=2/3VCC the 555

detects this through pin 6 and changes its output

LOW.

Then pin 7, the discharge pin, discharges C2 so that the circuit is ready for the next trigger.

So, Trigger→C2 charges→VC=2/3VCC→Alarm OFF
In short

C2 is the timing capacitor of the 555 monostable. Together with R4, it determines the duration for which the alarm stays ON.

3) C3 = 0.01 μF = 10 nF: noise filtering capacitor

C3 is connected between **pin 5 of the 555 and ground**.

Pin 5 is called: Control Voltage Pin.

Internally, the 555 normally uses reference levels around 1/3Vcc AND 2/3 Vcc.

Noise entering pin 5 can disturb these reference voltages. That can cause:

- accidental triggering
- unstable timing
- alarm jitter
- false alarms

C3 bypasses high-frequency electrical noise to ground.

So , Noise→C3→GND

Instead of, Noise→555 comparators

In short,

C3 is a noise bypass capacitor connected to pin 5. It stabilizes the internal reference voltage of the 555 and prevents false triggering due to electrical noise.

4) C4 = 0.22 μF: LDR sensor stabilization/filter capacitor

C4 is shown near the **LDR/transistor sensing section** of your circuit. It is associated with the transistor input network rather than the 555 timing stage. Its main purpose is to **smooth fast changes and noise from the LDR sensing circuit**. An LDR can respond to unwanted disturbances. Without filtering, these disturbances could make the transistor switch unnecessarily.C4 stores some charge and prevents the transistor's base voltage from changing instantaneously.

So instead of, tiny light disturbance→instant transistor switching

We get approximately, tiny disturbance→C4 absorbs/smooths it→no false trigger

It effectively forms an **RC low-pass filter** with the surrounding resistance.

In short,

C4 acts as a smoothing or noise-filtering capacitor in the LDR sensing stage. It reduces false triggering caused by rapid light fluctuations or electrical noise.

SUMMARY:

C1=TRIGGER

C2=TIMER

C3=555 NOISE FILTER

C4=LDR NOISE FILTER

WHY DID WE USE 2 POLAR AND 2 NON POLAR CAPACITORS?

Polar capacitors are used mainly for larger capacitance and fixed DC polarity, while nonpolar capacitors are used for smaller values and signal/noise filtering where polarity is not important.

large timing/coupling values → electrolytic →

polar small filtering/coupling values →

ceramic or film → non-polar.

COULD WE USE ALL NON -POLAR capacitors?

Yes *technically*, non-polar capacitors can be used if the capacitance and voltage ratings are correct. But for large values like 100 µF, polarized electrolytic capacitors are preferred because they are smaller, cheaper, and easier to find.

Engineering Design Considerations:

- LDR spacing is selected to ensure that the beam interruption reliably distinguishes human-sized intruders from smaller animals or airborne debris.
- The monostable timing components (R4, C2) determine the alarm duration, chosen to be long enough to provide adequate warning while avoiding unnecessary battery drain during unattended operation.
- Opaque tubing surrounding each LDR minimizes interference from ambient light, ensuring the sensors respond primarily to the laser beam.
- The two-transistor switching stage electrically isolates the sensitive LDR sensing network from the 555's trigger input, ensuring clean and reliable switching action.
- Component tolerances, particularly LDR resistance variation, may require minor adjustment of resistor values during testing to achieve reliable threshold behavior.

Expected Outcome:

A working breadboard prototype that reliably sounds an alarm for a set duration when the laser beam is interrupted by a human-sized object, then automatically resets without manual intervention, while ignoring smaller disturbances like insects or debris